SQL Mock Question Set – JOIN3_R_2 Pattern

Question 1 – Inventory Database

M2 Mock 1: Products With Total Quantity Purchased And Supplier Names

🗄️ Database Description

The `Inventory` database is designed to manage product inventory and purchasing activities. It contains information about products, suppliers, purchase orders, and individual products purchased through those orders.

The database supports analysis of product procurement, supplier relationships, purchasing quantities, and inventory-related activities.

📌 Problem Statement

The inventory management team wants a report that shows **each product, the total quantity purchased for that product, and the names of suppliers who supplied it**.

This report will help management analyze product procurement volumes and understand which suppliers are associated with each product.

📌 Relevant Tables

Products

Column	Description
`product_id`	Unique product identifier
`product_name`	Name of the product
`category`	Product category
`supplier_id`	Supplier associated with the product

`price`	Product price

Suppliers

Column	Description
`supplier_id`	Unique supplier identifier
`supplier_name`	Name of the supplier
`city`	Supplier location

Purchase_Orders

Column	Description
`order_id`	Unique purchase order
`supplier_id`	Supplier who fulfilled the order
`order_date`	Date of purchase order

Purchase_Items

Column	Description
`order_id`	Purchase order identifier
`product_id`	Product purchased
`quantity`	Quantity purchased

📌 Query to be Implemented

Write a SQL query to retrieve:

- `product_name`
- `total_quantity_purchased` — sum of all quantities purchased for the product

- `supplier_name`

Each row should represent a product and one of its associated suppliers.

📌 Expected Output

product_name	total_quantity_purchased	supplier_name
Product A	25	Supplier X
Product A	25	Supplier Y
Product B	40	Supplier X

Question 2 – SmartMart Database

M2 Mock 2: Products With Total Quantity Sold And Customer Names

🗄️ Database Description

The `SmartMart` database manages retail sales transactions and customer purchases. It contains information about customers, products, orders, and order-level product details.

The database can be used to analyze product sales, customer purchasing behavior, and order history.

📌 Problem Statement

The SmartMart management team wants a report showing **each product, the total quantity sold for that product, and the names of customers who purchased it**.

This report will help management identify product demand and understand the customer base associated with each product.

📌 Relevant Tables

Customers

Column	Description
`customer_id`	Unique customer identifier

`customer_name`	Name of the customer
`city`	Customer location

Products

Column	Description
`product_id`	Unique product identifier
`product_name`	Name of the product
`category`	Product category
`price`	Product price

Orders

Column	Description
`order_id`	Unique order identifier
`customer_id`	Customer who placed the order
`order_date`	Date of the order

Order_Items

Column	Description
`order_id`	Order identifier
`product_id`	Product purchased
`quantity`	Number of units purchased

📌 Query to be Implemented

Write a SQL query to retrieve:

- `product_name`
- `total_quantity_sold` — sum of all quantities sold for the product
- `customer_name`

Each row should represent a product and one of its purchasing customers.

📌 Expected Output

product_name	total_quantity_sold	customer_name
Laptop	15	John
Laptop	15	Sarah
Smartphone	22	John
Smartphone	22	David

Important: The total quantity should represent the **overall quantity sold for the product**, not the quantity purchased by the individual customer.

Question 3 – AdventureWorks Database

M2 Mock 3: Products With Total Quantity Ordered And Customer Names

🗄️ Database Description

The `AdventureWorks` database contains sales, product, customer, and person information for a retail/manufacturing business.

Sales transactions are stored through sales order headers and sales order details, while product and customer information is maintained in the corresponding product, customer, and person tables.

📌 Problem Statement

The sales analysis team wants a report showing **each product, the total quantity ordered for that product, and the names of customers who purchased it**.

This report will help the sales team understand product demand and identify the customers associated with each product.

📌 Relevant Tables

Production.Product

Column	Description
`ProductID`	Unique product identifier
`Name`	Product name
`ProductNumber`	Product number
`ListPrice`	Product selling price

Sales.SalesOrderDetail

Column	Description
`SalesOrderID`	Sales order identifier
`ProductID`	Product identifier
`OrderQty`	Quantity ordered

Sales.SalesOrderHeader

Column	Description
`SalesOrderID`	Sales order identifier
`CustomerID`	Customer identifier
`OrderDate`	Order date

Sales.Customer

Column	Description

`CustomerID`	Customer identifier
`PersonID`	Associated person identifier
`AccountNumber`	Customer account

Person.Person

Column	Description
`BusinessEntityID`	Person identifier
`FirstName`	Customer first name
`LastName`	Customer last name

📌 Query to be Implemented

Write a SQL query to retrieve:

- `product_name`
- `total_quantity_ordered` — total `OrderQty` for the product
- `customer_name` — customer's first name and last name

Each row should represent a product and one of its purchasing customers.

📌 Expected Output

product_name	total_quantity_ordered	customer_name
Mountain Bike	125	John Smith
Mountain Bike	125	David Brown
Road Bike	86	Sarah Jones
Road Bike	86	Michael Lee

Note: The total quantity should be calculated at the **product level** and then associated with its purchasing customers.

Question 4 – Student Database

M2 Mock 4: Courses With Total Enrollments And Student Names

🗄️ Database Description

The `Student` database manages academic information related to students, courses, enrollments, and instructors.

The database can be used to analyze student participation, course popularity, enrollment patterns, and academic relationships.

📌 Problem Statement

The academic administration team wants a report showing **each course, the total number of students enrolled in that course, and the names of students enrolled in it**.

This report will help the administration identify popular courses and understand which students are participating in each course.

📌 Relevant Tables

Students

Column	Description
`student_id`	Unique student identifier
`student_name`	Name of the student
`email`	Student email

Courses

Column	Description

`course_id`	Unique course identifier
`course_name`	Name of the course
`credits`	Course credits

Enrollments

Column	Description
`enrollment_id`	Unique enrollment identifier
`student_id`	Student identifier
`course_id`	Course identifier
`enrollment_date`	Date of enrollment

📌 Query to be Implemented

Write a SQL query to retrieve:

- `course_name`
- `total_enrollments` — total number of students enrolled in the course
- `student_name`

Each row should represent a course and one of its enrolled students.

📌 Expected Output

course_name	total_enrollments	student_name
Python Programming	30	John
Python Programming	30	Sarah
Python Programming	30	David
Database Systems	25	Michael
Database Systems	25	Priya

Important: `total_enrollments` must represent the **overall enrollment count for the course**, rather than a count calculated separately for each student.